Rafay Mahmood

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Education

Carnegie Mellon University — Expected May 2027
M.S. in Mechanical Engineering | GPA: 3.84/4.00 — *Pittsburgh, PA*
Thesis: Exoskeleton Control and State Estimation | Coursework: Robot Learning, Optimal Control & RL, Modern Control Theory, Deep Learning

The Pennsylvania State University — May 2025
B.S. in Mechanical Engineering | GPA: 3.49/4.00 — *University Park, PA*

Research and Professional Experience

Graduate Exoskeleton Researcher — September 2025 – Present
MetaMobility Lab - Carnegie Mellon University — *Pittsburgh, PA*

- Built a real-time, physics-based human hip torque estimator (nonlinear disturbance observer) on an NVIDIA Jetson from four IMUs, with closed-form dynamics 178x faster than the numerical model and no training data
- Leading its validation study against OpenSim inverse dynamics from motion capture across walking speeds, slopes, speed changes, and running (88 trials, 6 subjects), benchmarking it against an ML estimator
- Brought up the sensing pipeline: built the IMU and I^2C mux integration board, raised IMU read rate 1.8x with grouped register reads, and cut estimator latency from ∼115 to 68 ms
- Diagnosed I^2C bus faults and drifting BNO055 onboard fusion, becoming the lab's go-to for BNO055 debugging; built an error budget showing sensing is only ∼20% of estimator error
- Prototyped a small learned correction (∼3k weights, 8.5 μs per sample) on the estimator's outputs that lowered error in every trial under leave-one-subject-out testing

Undergraduate KINARM Robot Researcher — August 2024 – May 2025
Sensorimotor Neuroscience and Learning Lab - The Pennsylvania State University — *University Park, PA*

- Developed a nonlinear Simulink neuromuscular model combining vestibular, proprioceptive, and saccadic eye-movement feedback with an inverted-pendulum controller to simulate responses to mechanical perturbations and generate synthetic data for comparison with KINARM data

eVinci Component Design Intern — May 2024 – August 2024
Westinghouse Electric Company — *Cranberry, PA*

- Designed encapsulation segmentation and heat puck models in SolidWorks and Autodesk Inventor Pro, and updated the instrumentation and shielding design document 2 months ahead of internal design review

Projects

Golf Shot Optimization | *Python, MuJoCo, CMA-ES* — Feb 2026 – May 2026

- Built a MuJoCo golf simulation and CMA-ES optimizer that holed shots on two courses within 144 rollouts and cut a hazard course from 3 to 2 strokes via full-horizon sequence planning

Vehicle and Quadrotor Control | *Python, Webots, LQR, MRAC, A** — October 2025 – December 2025

- Designed an integral-augmented LQR lateral controller with PID speed control and A* planning to overtake a slower simulated vehicle, holding 0.50 m mean cross-track error over a full track loop
- Designed an MRAC controller that kept a simulated quadrotor tracking altitude commands through a 67.2% motor thrust loss that crashed the LQR baseline

Controls Education with Maglev | *C++, PID Control, Simulink, SolidWorks* — October 2025 – December 2025

- Led PID controller programming for an IR-sensed maglev system, tuning gains for stable position control from a nonlinear Simulink model of the magnet dynamics, and iterated housing designs in SolidWorks

Technical Skills

Controls: PID, LQR, MPC, Adaptive Control (MRAC), Nonlinear Control, State Estimation (NDO, VQF, Madgwick), Rigid-Body Dynamics, Path Planning (A*), Trajectory Optimization (CMA-ES)
Programming: Python, MATLAB, C++
Robotics & ML: PyTorch, MuJoCo, Webots, Simulink, NumPy, pandas, OpenSim
Embedded: Linux, NVIDIA Jetson, I2C/CAN
Software: SolidWorks, Autodesk Inventor, KiCad, Git, Vicon Motion Capture